
Outgoing Calls Forecasting

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1 PROBLEM DESCRIPTION

The dataset consists of information from 15300 customers of a telecommunication company, who had an active contract in the 9 months preceding the study. For 10000 of these customers you know the total number of outgoing calls in month 9 [denoted as y], and you have additional input variables describing several features of each customer's behavior. For the other 5300 customers, you have only information on the inputs and not on their outgoing calls in month 9.

Your goal is to predict y for the held-out 5300 customers.

There are 88 input variables, which are described below.

- **Characteristic variables of customer and of company contract.**
 - **tariff**: customer tariff plan [5 possible plans]
 - **payment**: method of payment [3 possible methods]
 - **gender**: gender [M-male, F-female, B-company]
 - **age**: customer's age in years
 - **area**: geographical activation zone [4 possible zones]
 - **channel**: channel of activation [8 possible channels]
 - **added.service.1**: presence of a first value-added service [N-no, Y-yes]
 - **added.service.2**: presence of a second value-added service [N-no, Y-yes]
- **Variables for the customer's traffic in the 8 months preceding the 9th. For each month $j = 1, \dots, 8$, the following variables are available.**
 - **m0j.out.peak.count**: number of outgoing calls at peak tariff times in month j
 - **m0j.out.peak.duration**: total duration of outgoing calls at peak tariff times in month j
 - **m0j.out.peak.value**: total value of outgoing calls at peak tariff times in month j
 - **m0j.out.offpeak.count**: number of outgoing calls at off-peak tariff times in month j
 - **m0j.out.offpeak.duration**: total duration of outgoing calls at off-peak tariff times in month j
 - **m0j.out.offpeak.value**: total value of outgoing calls at off-peak tariff times in month j

- `m0j.in.count`: number of incoming calls in month `j`
- `m0j.in.duration`: total duration of incoming calls in month `j`
- `m0j.sms.count`: number of SMS sent in month `j`
- `m0j.custom.service.count`: number of calls to Customer Services in month `j`

As already mentioned, the above variable are available for each month $j = 1, \dots, 8$.

Note: the variable `w` refers to weights and hence can be discarded.